

★ WEEK 6 · PART 1 · PRACTICE

The VAR Model, Worked by Hand

Forecast Two Series Jointly, Then Check It Won't Explode

A bivariate VAR(1), its stability, and its lag order

Part 1's core model, with the arithmetic shown. We write out a bivariate VAR(1) and forecast both series together - each borrowing the other's history - then check stability through the eigenvalues and pick the lag order with AIC and BIC.

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PART 1 · APPLICATIONS

Two Series, One Joint Model

Case A: a bivariate VAR(1) - write the two equations, then forecast both series one and two steps ahead, watching the cross-dynamics at work. Case B: check stability through the eigenvalues and choose the lag order with AIC and BIC. Then a Claude Code lab.

What Part 1 Told Us

Now we apply it - with numbers

1

A VAR models series together

Each variable depends on the recent past of every variable - so feedback, lead-lag, and co-movement are all captured at once.

2

The coefficient matrix holds the cross-dynamics

Its off-diagonal entries are how one series influences another. Forecasting is one matrix-vector multiply per step.

3

Stability and lag order come first

The eigenvalues must sit inside the unit circle, and the lag order p balances fit against a blow-up in parameters.

Everything here is a calculator exercise: one forecast, one eigenvalue check, one lag choice.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from the coefficient matrix.
- Forecast, or check stability and lag.
- Then read what the numbers say.

What you should watch

- How the cross term feeds one series into another.
- How the eigenvalues decide stability.
- How AIC and BIC can disagree.

A Bivariate VAR(1)

Forecast two returns together

A stock's monthly return and its sector's return move together. A VAR(1) captures that by letting each depend on both series' last values. Given the coefficient matrix and last month's returns, we forecast both one and two steps ahead - and see the cross term carry the sector's move into the stock's forecast

The setup

- Two returns: a stock and its sector.
- Last month: stock 2.0%, sector 1.0%.
- A given coefficient matrix and constant.

What we'll do by hand

- Write the two equations out.
- Forecast one step with a matrix multiply.
- Iterate for the two-step forecast.

The VAR(1), Written Out

Each equation borrows the other's past

$$\begin{aligned} y_{1,t} &= 0.2 + 0.5 y_{1,t-1} + \underbrace{0.3 y_{2,t-1}}_{\text{cross term}} + \varepsilon_{1,t} \\ y_{2,t} &= 0.1 + 0.1 y_{1,t-1} + 0.4 y_{2,t-1} + \varepsilon_{2,t} \end{aligned}$$

Each series depends on both series' last values. The cross term - 0.3 times the sector in the stock's equation - is where the multivariate payoff lives: the stock's forecast uses the sector's history.

The One-Step Forecast

One matrix-vector multiply

$$\hat{y}_t = c + A y_{t-1} = \begin{bmatrix} 0.2 \\ 0.1 \end{bmatrix} + \begin{bmatrix} 0.5 & 0.3 \\ 0.1 & 0.4 \end{bmatrix} \begin{bmatrix} 2.0 \\ 1.0 \end{bmatrix} = \begin{bmatrix} 1.50 \\ 0.70 \end{bmatrix}$$

The stock

- $0.2 + 0.5(2.0) + 0.3(1.0)$.
- Its own momentum plus the sector's pull.
- Forecast: 1.50%.

The sector

- $0.1 + 0.1(2.0) + 0.4(1.0)$.
- A little from the stock, more from itself.
- Forecast: 0.70%.

The Two-Step Forecast

Feed the forecast back in

$$\hat{y}_{t+1} = c + A \hat{y}_t : \quad \hat{y}_1 = 0.2 + 0.5(1.50) + 0.3(0.70) = \mathbf{1.16}, \quad \hat{y}_2 = \mathbf{0.53}$$

How it works

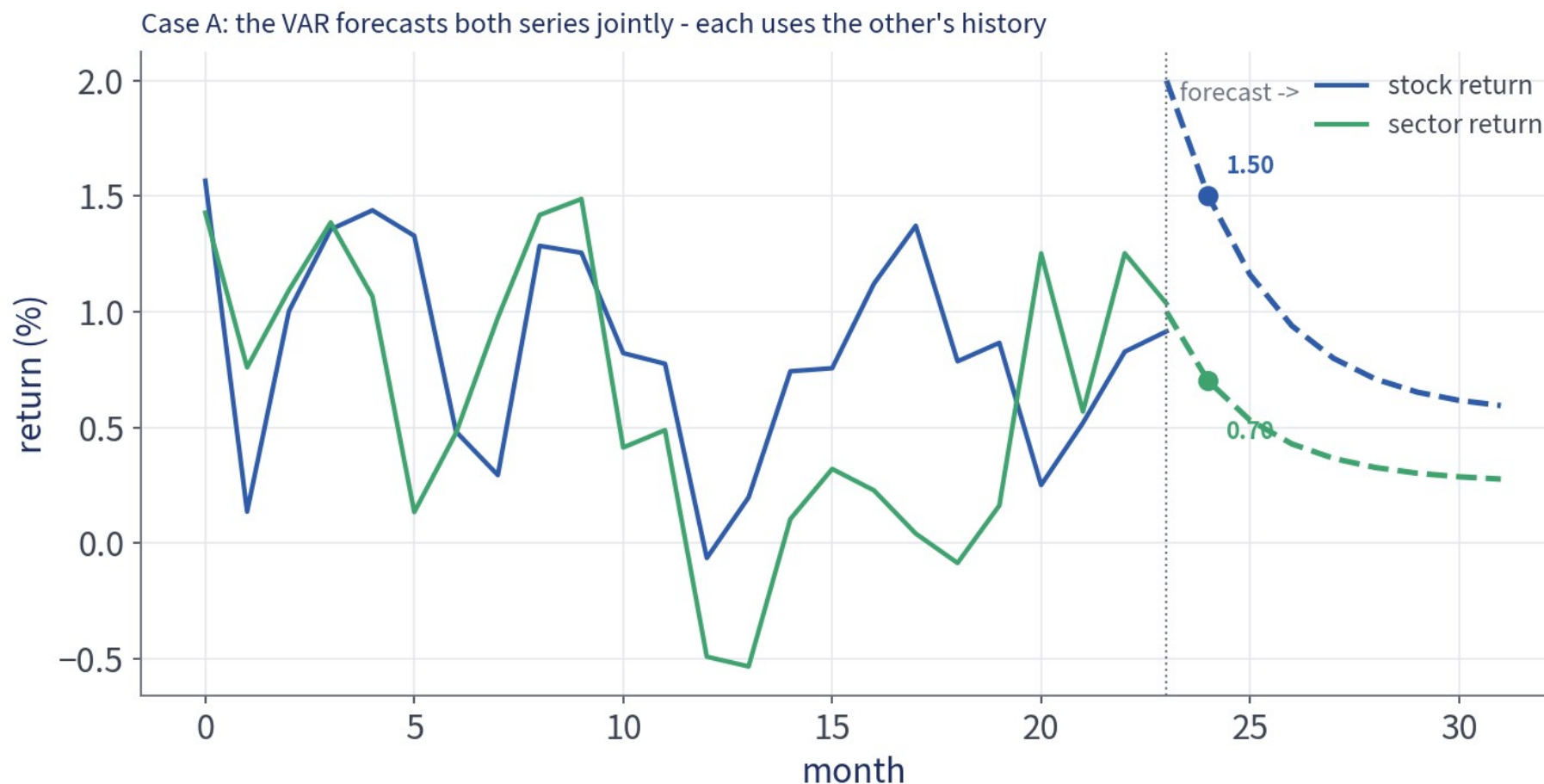
- Use the one-step forecast as the new input.
- Multiply by the matrix again.
- Add the constant.

The result

- Stock: 1.16%, sector: 0.53%.
- Both drift toward their long-run means.
- Because the VAR is stable.

Forecasting Both at Once

The stock and sector returns are modeled jointly, so each forecast draws on the other's history. Past the dotted line, the VAR projects both series forward together - and because it's stable, they settle toward their long-run means rather than wandering off



Why Joint Beats Isolated

The cross term earns its keep

What one-at-a-time misses

- The stock's move often follows the sector's.
- A univariate model ignores that lead-lag.
- It leaves information on the table.

What the VAR captures

- Feedback between the two series.
- Lead-lag through the cross terms.
- A joint forecast, internally consistent.

Stability and Lag Order

Two checks before you trust a VAR

A VAR is only useful if it doesn't explode and isn't overfit. Stability comes from the eigenvalues of the coefficient matrix: all must sit inside the unit circle. And the lag order p is chosen by information criteria - where AIC and BIC often disagree, BIC favoring the simpler model. We check both by hand

The two checks

- Stability: eigenvalues inside the unit circle.
- Lag order: minimize AIC or BIC.
- Both guard the forecast.

What we'll do by hand

- Compute the eigenvalues of the matrix.
- Confirm both are below one.
- Compare AIC and BIC across lags.

The Stability Condition

The multivariate unit-root test

stable $\iff$ all eigenvalues of A have $|\lambda| < 1$ (the multivariate unit-root condition)

The condition

- Look at the eigenvalues of the matrix.
- Each must have modulus below one.
- Then shocks fade and forecasts converge.

What failure looks like

- An eigenvalue at or above one.
- Shocks amplify instead of decay.
- Forecasts explode - the model is unusable.

The Eigenvalues

Both inside the unit circle

$$\lambda = \frac{\text{tr} \pm \sqrt{\text{tr}^2 - 4 \det}}{2} = \frac{0.9 \pm \sqrt{0.13}}{2} = \mathbf{0.63, 0.27} \text{ (both } < 1 \Rightarrow \text{stable)}$$

The calculation

- Trace 0.9, determinant 0.17.
- Eigenvalues 0.63 and 0.27.
- Both comfortably below one.

The contrast

- A matrix with entries 0.9 and 0.4 gives 1.25.
- Above one - that VAR would explode.
- Small coefficients keep it stable.

Choosing the Lag Order

Where AIC and BIC disagree

lag p	AIC	BIC	note
1	-8.20	-8.05	BIC's choice
2	-8.28	-8.02	AIC's choice
3	-8.25	-7.88	both worse

AIC picks the lowest value at $p=2$; BIC, with its heavier penalty on parameters, picks $p=1$. When they disagree, BIC's simpler model usually forecasts more reliably out of sample.

Inside the Circle Is Safe

The eigenvalues, plotted

Reading the picture

- The navy ring is the unit circle.



Every eigenvalue must land inside the ring

- for the VAR's forecasts to converge.

Guard the Forecast

Stability and parsimony together

Stability matters because

- An unstable VAR gives exploding forecasts.
- Shocks never die out.
- Always check the eigenvalues first.

Lag order matters because

- Too many lags overfit and forecast worse.
- Too few miss real dynamics.
- Let BIC lean you toward simplicity.

The VAR Routine

Check the model against the forecast

The whole workflow is short. Confirm the series belong together, pick the lag order with an information criterion, estimate each equation by ordinary least squares, and check stability through the eigenvalues. Then forecasting is just repeated matrix multiplication. The cross terms are what make it worth the trouble -

The forecast (Case A)

- One matrix multiply per step.
- Stock 1.50%, sector 0.70% next month.
- Each borrows the other's history.

The checks (Case B)

- Eigenvalues 0.63 and 0.27 - stable.
- BIC prefers $p=1$.
- Guarded and parsimonious.

Reproduce the Forecast and the Checks

Direct Claude Code to check your arithmetic

GOAL Forecast the bivariate VAR, then verify stability and lag selection.

[INPUT] Prompt · Claude Code

```
Write and run part1_practice.py that:  
CASE A - forecast:  
  * with  $c=[0.2,0.1]$ ,  $A=[[0.5,0.3],[0.1,0.4]]$ , and  
     $y=[2.0,1.0]$ , print the 1-step and 2-step forecasts.  
CASE B - stability and lag:  
  * print the eigenvalues of A and confirm  $|\lambda|<1$ ;  
  * show  $A'=[[0.9,0.4],[0.3,0.9]]$  is unstable;  
  * fit a VAR at  $p=1,2,3$  and print AIC and BIC.  
Then plot the joint forecast and the eigenvalues in  
the unit circle.
```

Check against the slides

- 1-step $[1.50, 0.70]$.
- 2-step $[1.16, 0.53]$.
- eigenvalues 0.63, 0.27.
- AIC picks $p=2$, BIC $p=1$.

Deliverable: part1_practice.py whose printed numbers match the forecasts and the checks.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Where do a VAR's cross-dynamics live?

In the off-diagonal coefficients of the matrix

Q2 When is a VAR stable?

When every eigenvalue sits inside the unit circle

Q3 Why might you follow BIC over AIC?

It favors the simpler, more robust model

All three come from the VAR forecast, the eigenvalue check, and the lag selection.

Three Pitfalls to Avoid

Where VAR modeling trips people

Common mistakes

- "Skip the stability check and just forecast."
- "More lags always fit better."
- "Model each series on its own."

The reality

- An unstable VAR explodes - check eigenvalues first.
- Extra lags overfit; let BIC discipline p .
- The cross terms are the whole point of a VAR.

Two Calculations to Remember

Part 1 applications in one screen

1

Forecast two series with one matrix multiply

From last month's $[2.0, 1.0]$ and the coefficient matrix, the one-step forecast was $[1.50, 0.70]$ and the two-step $[1.16, 0.53]$. The cross term carried the sector's move into the stock's forecast - the payoff of modeling jointly.

2

Check stability, then pick the lag

The eigenvalues of the matrix were 0.63 and 0.27 - both inside the unit circle, so the VAR is stable and its forecasts converge. Across lags, AIC picked $p=2$ and BIC $p=1$; when they disagree, BIC's simpler model usually travels better.

3

The routine is short and repeatable

Pick the lag, estimate each equation by least squares, confirm stability, then forecast by iterating the matrix. The cross-dynamics are what make the whole exercise worthwhile.

Next — Part 2 practice: Granger causality, testing whether one series really helps predict another.